



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2025-26
PRE BOARD EXAMINATION

CLASS: X
SUBJECT: BIOLOGY [SET A]

FULL MARKS: 80
TIME: 2 HOURS

*Answers to this Paper must be written on the paper provided separately.
You will not be allowed to write during the first 15 minutes.
This time is to be spent in reading the Question Paper.
The time given at the head of this Paper is the time allowed for writing the answers.*

*Section A is compulsory. Attempt any four questions from Section B.
The intended marks for questions or parts of questions are given in brackets [].
This question paper consists of seven printed pages.*

SECTION A
(Attempt all questions from this Section)

Question 1.

Choose the correct answer to the questions from the given options. (Do not copy the question, write the correct answer only):

[15]

- (i) Class 10 students went to a hospital visit and there they met a doctor. He asked them about the structural and functional unit of excretory system. One of them had answered him.

Identify from the options given below which is the correct option.

- (a) Nephron (b) Perikaryon
(c) Neuron (d) Axon

- (ii) Identify the type of tropism depicted in the diagram given below:



- (a) Geotropism (b) Thigmotropism
(c) Hydrotropism (d) Phototropism

- (iii) Riddhi tried to understand the function of two blood vessels in human body one which carries blood from the heart to lungs and the other from the lungs to the heart. She tabulated the pairs as follows:

Blood vessel	Function
P	Carries oxygenated blood
Q	Carries deoxygenated blood

- (a) P- Pulmonary Artery and Q- Pulmonary Vein
(b) P- Posterior Vena cava and Q- Renal Artery
(c) P- Pulmonary Vein and Q- Pulmonary Artery
(d) P- Renal Vein and Q- Renal Artery

- (iv) Shyam was trying to cross the road. He looked to his right and then to his left and then he crossed the road. Identify the parts of the human body associated with this act.
- Skeletal muscles, Hand, Ear
 - Eye, Nervous System and Skeletal Muscles
 - Spinal Cord, Smooth muscles and Eye
 - Hand, Spinal Cord and Skeletal Muscles
- (v) Assertion (A): Active transport does not require ATP for the absorption of minerals by roots.
Reason (R): Active transport involves the movement of minerals against their concentration gradient.
- A is true and R is false
 - Both A and R are true
 - A is false and R is true
 - Both A and R are false
- (vi) Akshita, a caring citizen, is researching on the impact of air pollution on human health. She with her friends were trying to find out which of the following pollutants in the air are contributing to respiratory diseases such as asthma.
Identify who had given the correct answer:
- Rina- CO₂ and CFCs
 - Akshita- CO₂ and Water vapour
 - Sohini- CO₂ and CO
 - Alam- DDT and Oil
- Rina is correct
 - Alam is correct
 - Sohini is correct
 - Akshita is correct
- (vii) If a pure purple flower (WW) plant is crossed with a pure white flower plant (ww), the phenotypic ratio (P) and genotypic ratio (Q) of F₂ generation will be:
- P- All Purple flower and Q- All Ww
 - P- 1 Purple flower: 3 White flower and Q- 1WW: 2Ww: 1ww
 - P- 3 Purple flower: 1 White flower and Q- 1 WW: 2 Ww: 1ww
 - P- 2 Purple flower: 1 White flower and Q- 2 WW: 1 ww
- (viii) Ram observed that on a landfill few syringes and dirty dressings were dumped. He was trying to segregate the different kinds of soil pollutants. Help him identify to which category does syringes and dirty dressings belong:
- | | |
|-----------------------|---------------------|
| P. Domestic waste | Q. Biomedical waste |
| R. Agricultural waste | S. Industrial waste |
- Only P
 - Only R
 - Only S
 - Only Q
- (ix) Assertion (A) – Darwin’s theory suggests that organisms which can adapt fast to the changes are more likely to survive and reproduce.
Reason (R) – Darwin’s main principle was based on ‘Use and Disuse’ of body parts.
- A is true and R is false
 - Both A and R are true
 - A is false and R is true
 - Both A and R are false

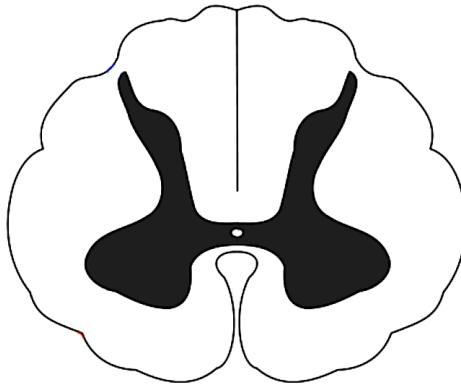
- (x) Sayani was a mother of two kids. She and her husband thought of family planning and they decided to go for an operative method to control child birth. Identify the process which Sayani will go through from the following options:
- (a) Ovectomy (b) Vasectomy
(c) Tubectomy (d) Hysterectomy
- (xi) Rapid population growth in India has led to increased pressure on ____P____, resulting in challenges like inadequate ____Q____ for the growing population. Identify P and Q:
- (a) P- Natural resources and Q- Food
(b) P- Technological Advancements and Q- Water
(c) P- Rural areas and Q- Source of entertainment
(d) P- Cultural heritage and Q- Unemployment
- (xii) There are certain norms which have been laid down under the title Euro/ Bharat norms that are applicable to automobiles. These aim to effectively cut down Sulphur and Nitrogen oxides from automobile exhausts. BS emission norms which were enforced for the entire country on April 1st 2017:
- (a) BS I (b) BS II
(c) BS III (d) BS IV
- (xiii) A 35 year old woman possesses some symptoms of Diabetes mellitus. Identify the symptom/s which is seen in her due to this disease:
- P- Excessive thirst. R- Weight Loss
Q- Weight gain. S- Feeling energetic
- (a) P and R (b) Q and R
(c) Q and P (d) Q and S
- (xiv) Ravi was studying about the different sequence of events in a woman. Identify the correct sequence of the events from the following options:
- (a) Parturition, Ovulation, Fertilisation, Implantation
(b) Ovulation, Parturition, Fertilisation, Implantation
(c) Ovulation, Parturition, Implantation, Fertilisation
(d) Ovulation, Fertilisation, Implantation, Parturition
- (xv) Assertion (A) – Anaphase is the shortest phase in cell division.
Reason (R) – Chromosomes align in the equatorial plane of the cell during anaphase.
- (a) A is true and R is false (b) A is false and R is true
(c) Both A and R are true (d) Both A and R are false.

Question 2

- (i) Give the biological or technical terms for the following: [5]
- (a) Enzyme present in tear.
(b) A condition in which the lens of the eye turns opaque.
(c) The growth movement of plant parts in response to chemical stimulus.
(d) An instrument used to measure the rate of transpiration.
(e) Initiator for the process of clotting of blood.

- (ii) Given below is a transverse section of the spinal cord. Read the information given below the diagram and fill in the blanks:

[5]



The spinal cord extends from the medulla oblongata of the brain and runs down to the lower part of the body. It conducts impulses from the skin and muscles to the brain and vice versa.

The spinal cord is a part of the ____ (a) ____ (Peripheral Nervous System/ Central Nervous System). It is covered by the ____ (b) ____ (Meninges/ Pericardium). It is protected by a bony structure called ____ (c) ____ (Sternum/ vertebral column). The grey matter in the spinal cord consists of ____ (d) ____ (cyton/ axon) and the white matter consists of ____ (e) ____ (cyton/axon).

- (iii) Choose the odd one out and write the categories to which the others belong:

[5]

- (a) Sub stomatal space, Lenticels, Palisade cell, Spongy cells.
- (b) Somatotropin, Prolactin, Oxytocin, Luteinizing hormone
- (c) Bowman's Capsule, Venule, Loop of Henle, PCT.
- (d) Retina, Auditory Nerve, Conjunctiva, Cornea.
- (e) Nucleosome, Chromatin, Corpus luteum, Histone.

- (iv) In a Biology surprise test, the teacher wrote few questions based on the topic Ear on the board and asked the students to solve them. Aayana was not being able to solve the questions. Help her in solving the following questions:

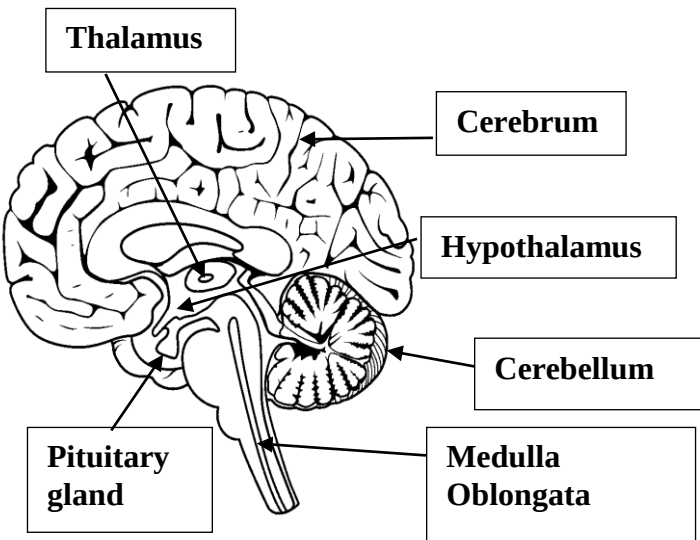
[5]

- (a) Name the smallest bone in our body.
- (b) Which part of the ear helps in maintaining the static balance of our body?
- (c) Name the tissue which is present in Pinna.
- (d) Mention the location of Organ of Corti.
- (e) Mention the function of Cochlea.

- (v) Given below is a cross section of human brain. Match the structures labelled with their correct functions:

[5]

Example: Thalamus- 2. Relays pain and pressure impulses to cerebrum

Cross section of Human Brain	Functions
	1. Controls the activities of internal organs like breathing. 2. Relays pain and pressure impulses to cerebrum. 3. Controls the balance of the body and muscular activities. 4. Releases tropic hormones. 5. Seat of intelligence. 6. Controls body temperature.

SECTION B

(Attempt any four questions from this section)

Question 3

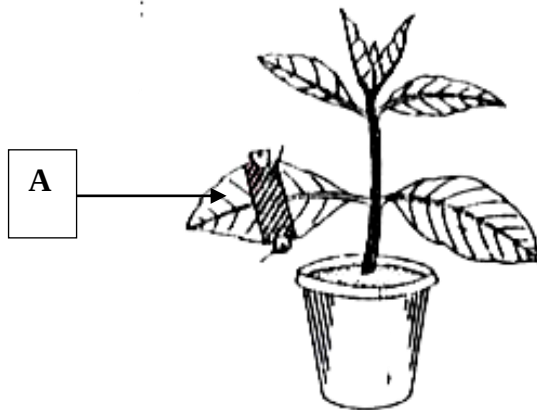
- Define Photosynthesis. [1]
- "Testes are present in the scrotum sac outside human male body". Explain why. Mention the function of Leydig Cells. [2]
- Ravi went to watch a movie with his friend. As he entered the dark hall, he was unable to observe anything for first few minutes. Slowly he could start observing things clearly. Mention the changes observed in the eyes of Ravi during this adaptation. [2]
- Kritika was trying to understand the different kinds of reflexes by relating with real life situations. Help Kritika identify the reflexes for the following: [2]
 - Knee-jerk
 - Giving a hand signal
- Draw a neat labelled diagram to show the correction of Hyperopia. [3]

Question 4

- Define Spermatogenesis. [1]
- Ultrafiltration is an important process to filter out the nitrogenous waste from our body. Explain the reason for Ultrafiltration to take place in Glomerulus. [2]
- Mendel while performing his experiment on tall and dwarf Pea plants had observed that tall was the character which was seen more in the offspring's and it suppressed the other character. Identify the Law which can be associated with this and state the Law. [2]
- Few students of class 10 for their Biology project had crossed a pure inflated pod pea plant with a pure constricted pod pea plant. They observed the F₁ generation and F₂ generation. What do you think would be the phenotype of F₁ generation offsprings and Phenotypic ratio of F₂ generation offsprings. [2]
- Draw a well labelled diagram of the structural and functional unit of nervous system. [3]

Question 5

- (i) Define the term Speciation. [1]
- (ii) Sabita was sprinkling DDT on the plants in her garden. She observed that after few hours, the plants wilted and they eventually died. The gardener told her that excessive DDT on the plants killed them. Explain with reasons why did the plants die due to excessive DDT. [2]
- (iii) Soham was studying about population explosion and its ill effects on resources. He observed that there are few major reasons for the population explosion in India. Explain two reasons for population explosion in India. [2]
- (iv) Differentiate between Karyotype and Karyokinesis on the basis of definition. [2]
- (v) Observe the diagram given below and answer the following questions: [3]



- (a) Mention the aim of the experiment.
- (b) What do you observe in leaf “A” at the end of the experiment when tested with Iodine solution?

Question 6

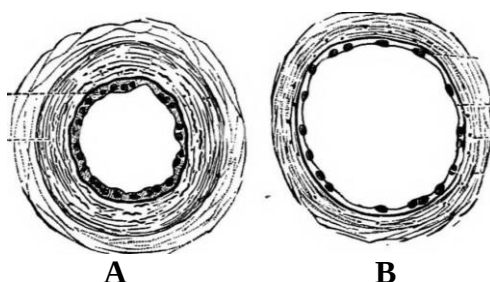
- (i) ‘Auxins play a very important role in Phototropism’. Explain the statement. [1]
- (ii) Hriday went to Indian Museum with his mother and observed a picture and a skull of the first man-like human ancestor. Identify the human ancestor and mention the cranial capacity of the ancestor. [2]
- (iii) The Pradhan of the village declared that on 2nd October everyone will participate in “Swachh Bharat Abhiyan”. He explained to the villagers the objectives of this programme. Mention two important objectives of this Abhiyan. [2]
- (iv) Soham was trying cross pollination between a pure Pea plant with axillary flower (AA) and a pure Pea plant with terminal flower (aa). Help Soham identify the gametes of F₂ generation and Genotypic ratio of F₂ generation offsprings. [2]
- (v) Adrija was trying to study the rate of transpiration in a broad leaf and a narrow leaf. She observed that the rate of transpiration was different in the two leaves. [3]
 - (a) Explain with reasons the cause for difference in transpiration in both the leaves.
 - (b) Xerophytes have leaves reduced to spines. Explain why.

Question 7

- (i) Define Demography. [1]
- (ii) In our body Synapse is an important part of nervous system. Define synapse. Mention the function of corpus callosum. [2]

- (iii) Study the diagram given below and differentiate between A and B on the basis of structural difference.

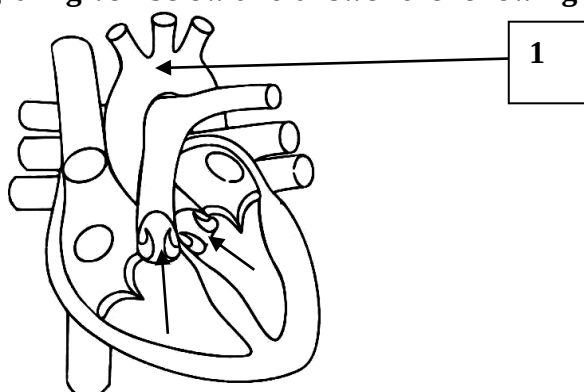
[2]



- (iv) Rohit was observing different phases of mitosis in the permanent slides in his lab. He was observing a stage which comes after metaphase. Mention two features of that phase.
- (v) Observe the diagram given below and answer the following questions.

[2]

[3]



- (a) Mention the function of the part labelled as “1”.
- (b) Which part of the heart is in the contraction phase? Give suitable reasons to justify your answer.

Question 8

- (i) Name the instrument used to measure geotropism.
- (ii) Rita was studying about Industrial Melanism and she came across an example of Industrial Melanism. Mention the scientific name of the organism and explain the effect of Industrial Melanism on the organism.
- (iii) Name the order to which human beings belong and mention the name of the French Biologist who proposed Inheritance of Acquired characters.
- (iv) Chromatin fibre is present inside the nucleus. It is coiled and super coiled as we see in a typical telephone cord. Mention the components which are present in chromatin fibre.
- (v) Observe the diagram given below and answer the following questions:

[1]

[2]

[2]

[2]

[3]



- (a) Identify the type of pollution shown in the above picture.
- (b) Mention two sources which causes this pollution.